

Burgers 2D — where the nonlinear manifold stops paying

trajectory rel-L2 (16 held-out trajectories)

intrinsic dim 6
(5 params + t)

10^{-1}

10^{-2}

2

4

6

8

12

16

24

32

64

latent dimension k

- coordinate ROM (full grid, weak64)
- coordinate ROM (NNLS-EQ, m=256)
- coord. inferred-latent floor (oracle)

- POD-LSPG (same solver)
- POD, same weak objective
- POD projection floor

coord. ROM best (K=16)